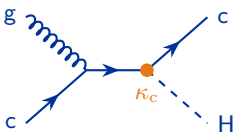
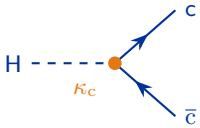
Constraints on Higgs-charm couplings at CMS



Chirayu Gupta
on behalf of the CMS Collaboration

Vrije Universiteit Brussel



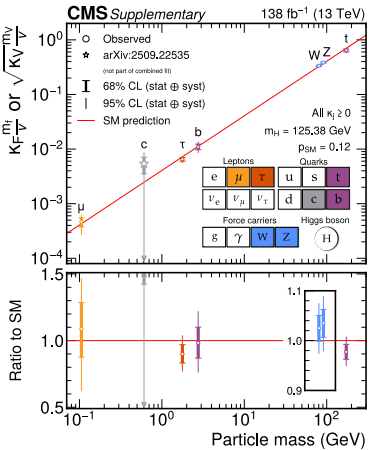
Higgs 2026, Louvain-la-Neuve, 28 October 2026





Why the Higgs-charm coupling?

- Since the 2012 discovery: couplings to t, b, τ measured, evidence for μ (JHEP 01 (2021) 148) $\rightarrow$ backup; all SM-like
- **2nd-generation quark Yukawas are the open frontier**; charm is the most accessible
  - $\mathcal{B}(\mathrm{H} \rightarrow \mathrm{c\bar{c}}) \approx 2.9\%$, about 20$\times$ below $\mathrm{H} \rightarrow \mathrm{b\bar{b}}$
  - huge QCD backgrounds; c jets are hard to tell from b and light jets
- $\kappa_\mathrm{c} = y_\mathrm{c}/y_\mathrm{c}^\mathrm{SM}$ (SM: $\kappa_\mathrm{c} = 1$)
- Flavour-compatible BSM (e.g. spontaneous flavour violation) allows $|\kappa_\mathrm{c}| \sim 3$: current sensitivity is entering this region
- Does the Higgs mechanism also give the 2nd-generation quarks their mass?



CMS, Rep. Prog. Phys. 89 (2026) 087801 (suppl.);
κ_c point: separate $\mathrm{t\bar{t}H}$+VH result, not in the fit

How CMS probes κ_c

Direct $H \to c\bar{c}$ (rate $\propto \kappa_c^2/\Gamma_H$) slides 4–7

$VH(H \to c\bar{c})$ PRL 131 (2023) 061801

$t\bar{t}H(H \to c\bar{c})$ + VH comb. PRL 136 (2026) 011801

Boosted $H \to c\bar{c}$ (mostly ggH) PRL 131 (2023) 041801

$t\bar{t}H$+**VH comb.:** $|\kappa_c| < 3.5\,(2.7)$

H + charm production (cg → cH) slides 8–10

$cH(H \to \gamma\gamma)$ JHEP 11 (2025) 060

$cH(H \to WW)$ + comb. JHEP 08 (2026) 051

comb.: $|\kappa_c| < 47\,(51)$

Indirect: H rates and width slide 11

On-shell $H \to 4\ell$ PRD 112 (2025) 112001

Off-shell + on-shell $H \to 4\ell$ **NEW** arXiv:2607.23352

$\kappa_c \in [-3.3, 2.8]$ **to** $[-8.6, 7.5]$ (by assumptions)

Rare exclusive decays slide 11

$H \to J/\psi\gamma,\ \psi(2S)\gamma$ PLB 865 (2025) 139462

sensitive to the ***sign*** of κ_c

$-166 < \kappa_c < 208$

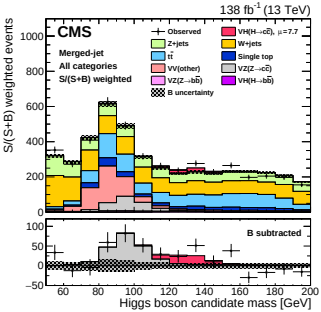
$\mu = \sigma\mathcal{B}/(\sigma\mathcal{B})_{SM}$, Γ_H = total Higgs width; μ limits and κ_c intervals at 95% CL: obs. (exp.)

All κ_c results: **Run 2** (13 TeV, up to 138 fb^{-1}); Run 3 so far: better triggers (slide 13). **NEW**: public in 2026

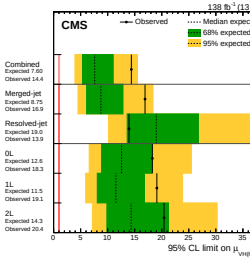
Direct $\mathrm{H} \to \mathrm{c\bar{c}}$: VH($\mathrm{H} \to \mathrm{c\bar{c}}$)

PRL 131 (2023) 061801, CMS-HIG-21-008

- $\mathrm{W/Z} \to$ 0, 1, 2 leptons; main backgrounds V+jets and $\mathrm{t\bar{t}}$, normalised in control regions
- **Resolved** ($p_{\mathrm{T}}^{\mathrm{H}} < 300\,\mathrm{GeV}$, >95% of signal): two DeepJet c-tagged jets; fit to a BDT
- **Merged** (large-R jet $p_{\mathrm{T}} > 300\,\mathrm{GeV}$): <5% of signal but less background; ParticleNet $\mathrm{c\bar{c}}$ tagging (3 purity categories) and mass regression; fit to $m(\mathrm{H_{cand}})$
- VZ($\mathrm{Z} \to \mathrm{c\bar{c}}$) observed at 5.7σ: validates the method



Merged-jet only; bottom: VZ (grey, ≈90 GeV) and VH (red) after background subtraction



μ limits per channel (inner band 68%)

VH($\mathrm{H} \to \mathrm{c\bar{c}}$)

95% CL obs. (exp.) $\to$ backup

$\mu < 14\,(7.6)$

$1.1 < |\kappa_{\mathrm{c}}| < 5.5 \quad (|\kappa_{\mathrm{c}}| < 3.4)$

The most sensitive single channel of Run 2 (expected $\mu < 7.6$); statistics-limited

Direct $\mathrm{H} \to \mathrm{c\bar{c}}$: boosted H (ggF-enriched)

PRL 131 (2023) 041801, CMS-HIG-21-012

Boosted regime ($p_{\mathrm{T}}^{\mathrm{H}} > 450$ GeV)

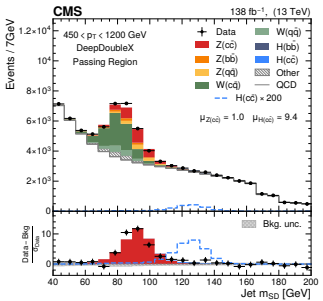
- High-p_{T} Higgs bosons to suppress QCD; signal 55% ggF, 25% VBF, 20% VH
- $\mathrm{c\bar{c}}$ reconstructed as one large-radius jet; soft drop removes soft, wide-angle radiation and sharpens the jet mass m_{SD}

DeepDoubleX tagger

- Deep neural network on low-level jet features: passing (signal-rich) and failing (background-rich) regions
- Decorrelated from m_{SD}, so the QCD background is estimated from data in the failing region

Statistical analysis

- Simultaneous binned fit of m_{SD} in all tagger regions; upper limit from the profile likelihood ratio
- Validation: **$\mathrm{Z} \to \mathrm{c\bar{c}}$ observed for the first time in Z+jets** at high p_{T}, well over 5σ



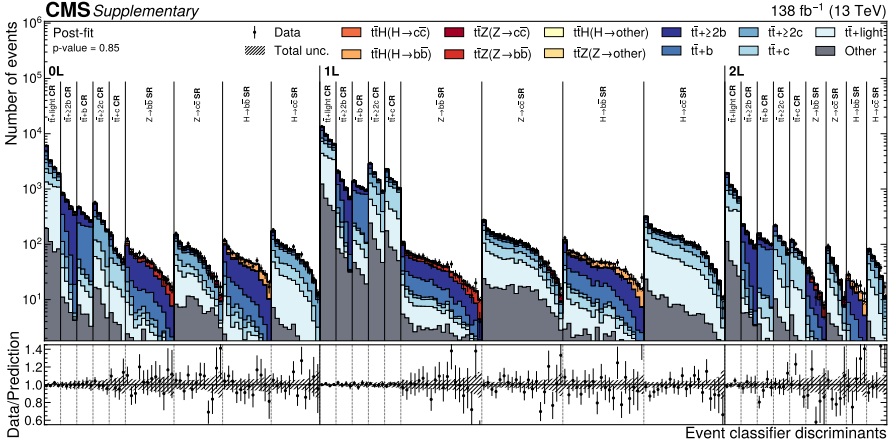
Passing region: peak is $\mathrm{Z} \to \mathrm{c\bar{c}}$, $\mathrm{H} \to \mathrm{c\bar{c}}$ dashed ($\times 200$)

Upper limit, 95% CL obs. (exp.)

$\sigma\mathcal{B}/(\sigma\mathcal{B})_{\mathrm{SM}} < 47\,(39)$

no κ_{c} interpretation in the paper

PRL 136 (2026) 011801, CMS-HIG-24-018 3 channels (0L, 1L, 2L) $\times$ {5 $t\bar{t}$+jets control regions + 4 signal regions}, fitted simultaneously



Post-fit ParT discriminants in all CRs and SRs; data agree with the fitted model (p-value 0.85)

One fit measures $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z(c\bar{c})$, $t\bar{t}Z(b\bar{b})$ and every $t\bar{t}$+jets flavour component

PRL 136 (2026) 011801, CMS-HIG-24-018

Strategy

- Simultaneous $t\bar{t}H(c\bar{c})$ and $t\bar{t}H(b\bar{b})$: same signature apart from the c/b-jet multiplicity; $t\bar{t}Z(c\bar{c})$, $t\bar{t}Z(b\bar{b})$ as standard candles in their own regions
- Every jet in one of 11 ParticleNet tagging categories (B0–B4, C0–C4, L0); ParT event classifier with 4 signal and 5 $t\bar{t}$+jets classes → backup

Backgrounds

- Leading: $t\bar{t}$+jets, split by extra-jet flavour (+light, +c, +≥2c, +b, +≥2b), each normalised in its own CR
- QCD multijet negligible, even in 0L: a dedicated classifier output removes it by ∼10^4, to <1% of $t\bar{t}$+jets

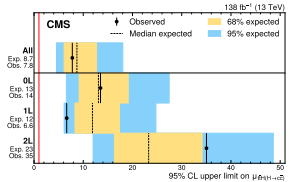
Regions (on $D_{t\bar{t}X}$, the summed signal score)

- Analysis region $D_{t\bar{t}X} > 0.6$: 4 SRs (> 0.85, by $t\bar{t}H/t\bar{t}Z$and $c\bar{c}/b\bar{b}$) and 5 CRs (0.6–0.85)
- Validation region 0.4–0.6 with the same SR/CR structure: good agreement → backup

$t\bar{t}H(H \to c\bar{c})$, 95% CL obs. (exp.)

$\mu < 7.8\,(8.7)$
$|\kappa_c| < 3.0\,(3.3)$ ($\kappa_b = 1$)
$|\kappa_c| < 4.2$ obs. (κ_b fitted) → backup

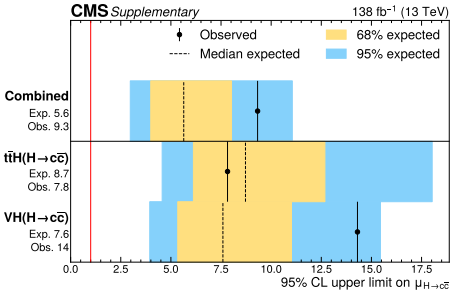
	best fit $\hat{\mu}$	obs. (exp.) signif.
$t\bar{t}H(b\bar{b})$	$0.91^{+0.26}_{-0.22}$	4.4σ (4.5σ)
$t\bar{t}H(c\bar{c})$	-1.6 ± 4.5	–
$t\bar{t}Z(b\bar{b})$	$1.47^{+0.45}_{-0.41}$	3.5σ (2.4σ)
$t\bar{t}Z(c\bar{c})$	$1.02^{+0.79}_{-0.84}$	1.2σ (1.3σ)



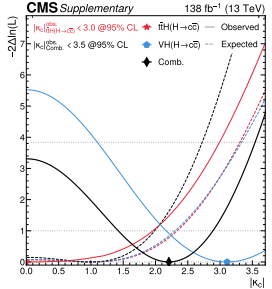
$\mu_{t\bar{t}H(c\bar{c})}$ limits per channel; yields → backup

Combination of $\mathrm{t\bar{t}H}$ and VH: best expected direct sensitivity to κ_c

PRL 136 (2026) 011801, CMS-HIG-24-018



$\mu_{\mathrm{H}\to\mathrm{c\bar{c}}}$ limits: $\mathrm{t\bar{t}H}$, VH, combined (inner band 68%)



$-2\Delta \ln L$ vs. $|\kappa_c|$: $\mathrm{t\bar{t}H}$, VH, comb.; solid obs., dashed exp. ($\kappa_b = 1$)

$\mathrm{t\bar{t}H}$ + VH

95% CL obs. (exp.)

$\mu_{\mathrm{H}\to\mathrm{c\bar{c}}} < 9.3$ (5.6)

$|\kappa_c| < 3.5$ (2.7)

SM production, $\kappa_b = 1$

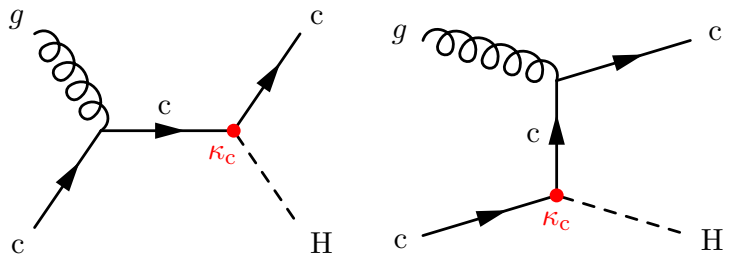
- Expected μ limit 26% better than VH alone (36% better than $\mathrm{t\bar{t}H}$ alone)
- $\mathrm{t\bar{t}H}$ alone (3.0) beats the combination (3.5) on observed because VH fluctuated up ($\hat{\mu} = 7.7$)

Expected $|\kappa_c| < 2.7$ (obs. 3.5): the expected reach is now below the BSM-motivated $|\kappa_c| \sim 3$

H + charm production: κ_c in production

JHEP 11 (2025) 060, CMS-HIG-23-010

- κ_c enters the **production** vertex, independent of the $H \to c\bar{c}$ decay: clean H decay + a c-tagged jet
- $\sigma_{cH} = 90\,\text{fb}$: $\sigma\mathcal{B} \approx 0.2\,\text{fb}$ ($\gamma\gamma$), $\approx 20\,\text{fb}$ (WW)
- Challenges: κ_c-**independent H+c** from $g \to c\bar{c}$ splitting in ggH (same final state); soft, often forward c jets; flavour-scheme modelling ($\to$ T. Bevilacqua, next talk)
- **"Flat direction"**: κ_c is varied while all other Higgs rates stay at their SM values, so only cH production changes



LO diagrams for cH production; κ_c enters at the marked vertices

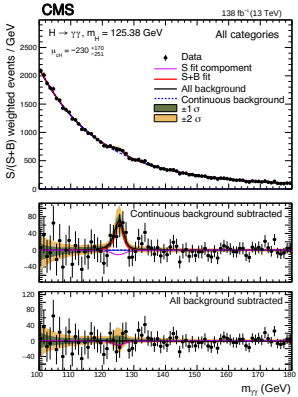
JHEP 11 (2025) 060, CMS-HIG-23-010

- First search for cH with H→ $\gamma\gamma$, 138 fb^{-1}
- Two BDTs, trained without c-tag inputs: cH vs. ggH (largest resonant background), cH vs. continuum $\gamma\gamma$+jets
- 9 categories per year (27 in total); signal from fits to $m_{\gamma\gamma}$
- Plot: the 125 GeV peak is fully explained by the other Higgs processes (mostly ggH); nothing is left for cH, best fit $\hat{\mu}_{\mathrm{cH}} = -230^{+170}_{-251}$

cH(H→ $\gamma\gamma$)

95% CL obs. (exp.)

$\mu_{\mathrm{cH}} < 243\,(355)$, $|\kappa_{\mathrm{c}}| < 38\,(73)$



$m_{\gamma\gamma}$, all categories, weighted by S/(S+B)

$|\kappa_{\mathrm{c}}| \lesssim 40$, **not ∼3:** $\sigma\mathcal{B} \approx 0.2$ **fb, and ggH H+c mimics the signal**

cH(H→WW) and the $\gamma\gamma$ + WW combination

- H→WW→e$\nu\mu\nu$ + c jet; signal regions with 1 or >1 c jet; two BDTs (vs. non-H and vs. other H processes)
- κ_c-independent H+c background fixed to SM, so μ_{cH} is the only parameter of interest (1 POI); floating it → backup
- WW alone: $\mu_{cH} < 1065\,(506)$, $|\kappa_c| < 211\,(95)$; post-fit → backup

$\gamma\gamma$ + WW combination (1 POI)

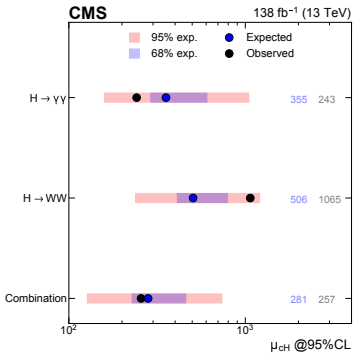
95% CL obs. (exp.)

$\mu_{cH} < 257\,(281)$, $|\kappa_c| < 47\,(51)$

obs. weaker than $\gamma\gamma$ alone (38): WW fluctuated up

($\hat{\mu}_{cH} = 390$ vs. -230 in $\gamma\gamma$)

JHEP 08 (2026) 051, CMS-HIG-24-009

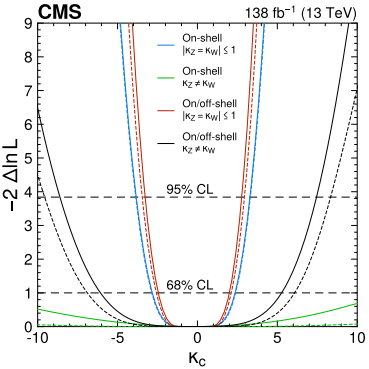


μ_{cH} limits: $\gamma\gamma$, WW, combined

κ_c from production alone: statistics-limited, independent of the $H \to c\bar{c}$ decay

Indirect constraints on κ_c

- Large κ_c adds $c\bar{c} \to H$ production and increases Γ_H: on-shell $H \to 4\ell$ rates constrain it
  κ_V (κ_Z, κ_W): HZZ, HWW coupling modifiers. Also (κ_b, κ_c) from the Higgs p_T spectrum → backup
- **On-shell $H \to 4\ell$**, $|\kappa_Z = \kappa_W| \leq 1$:
  $\kappa_c \in [-3.7, 3.2]$ ($[-3.8, 3.3]$)
- NEW **Off-shell + on-shell $H \to 4\ell$**: the off-shell rate breaks the κ_V–Γ_H degeneracy
  - $|\kappa_V| \leq 1$: $\kappa_c \in [-3.3, 2.8]$ ($[-3.5, 2.9]$)
  - κ_Z, κ_W free: $\kappa_c \in [-8.6, 7.5]$ ($[-9.6, 8.3]$)
  - width measurement: F. Errico's talk (Tue)
- $H \to J/\psi\gamma$: $-166 < \kappa_c < 208$ (J. Bierkens' talk, Thu)



arXiv:2607.23352 Fig. 8: blue on-shell and red on+off-shell with $|\kappa_Z = \kappa_W| \leq 1$; green on-shell and black on+off-shell with $\kappa_Z \neq \kappa_W$; solid obs., dashed exp. Text intervals: Table 3 (no custodial symmetry)

With $|\kappa_V| \leq 1$, $H \to 4\ell$ matches the direct searches, under different assumptions → backup

Where we stand: CMS constraints on κ_c

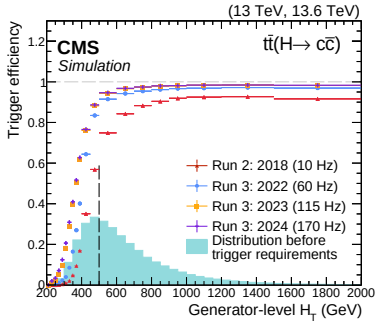
Probe	Reference	κ_c at 95% CL: obs. (exp.)	Main assumptions
$t\bar{t}H + VH(H \to c\bar{c})$	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.5$ (2.7)	$\kappa_b = 1$, SM production, no BSM decays
$t\bar{t}H(H \to c\bar{c})$	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.0$ (3.3)	same
$VH(H \to c\bar{c})$	PRL 131 (2023) 061801	$1.1 < \lvert\kappa_c\rvert < 5.5$ (3.4)	same
On+off-shell $H \to 4\ell$	arXiv:2607.23352	$[-3.3, 2.8]$ ($[-3.5, 2.9]$)	$\lvert\kappa_V\rvert \leq 1$, $\kappa_{b,t} = 1$, other light κ_q profiled, $\Gamma_{BSM} \geq 0$
On-shell $H \to 4\ell$	PRD 112 (2025) 112001	$[-3.7, 3.2]$ ($[-3.8, 3.3]$)	$\lvert\kappa_Z = \kappa_W\rvert \leq 1$, otherwise same
On+off-shell $H \to 4\ell$	arXiv:2607.23352	$[-8.6, 7.5]$ ($[-9.6, 8.3]$)	κ_Z, κ_W free, otherwise same
cH ($\gamma\gamma$ + WW)	JHEP 08 (2026) 051	$\lvert\kappa_c\rvert < 47$ (51)	flat direction, from μ_{cH} limit
$H \to J/\psi\gamma$	PLB 865 (2025) 139462	$[-166, 208]$ ($[-128, 169]$)	$\kappa_\gamma = 1$, $123\,\mathrm{fb}^{-1}$

The rows use different coupling models and are not directly comparable. $t\bar{t}H$ alone beats the combination on observed because VH fluctuated up; the combination has the best *expected* sensitivity (2.7).

Direct searches and off-shell + on-shell $H \to 4\ell$ both reach $\lvert\kappa_c\rvert \sim 3$, where flavour-compatible BSM scenarios live

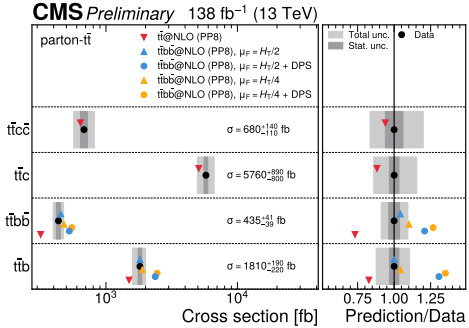
Toward Run 3: better triggers, measured backgrounds

Run 3: ML b/c tagging at the HLT



arXiv:2608.18331, simulation: $t\bar{t}H(H \to c\bar{c})$ trigger efficiency, Run 2 (2018) vs. Run 3 (2022–24), 13.6 TeV. HLT taggers: J. Daniel's talk (Mon) → backup

Run 2: $t\bar{t}H(H \to c\bar{c})$ background measured



CMS-PAS-TOP-26-003 **NEW**, Run 2 data (138 fb^{-1}):

$$\sigma_{t\bar{t}c\bar{c}} = 680 \pm 40\,(\text{stat})\,^{+130}_{-100}\,(\text{syst})\,\text{fb}$$

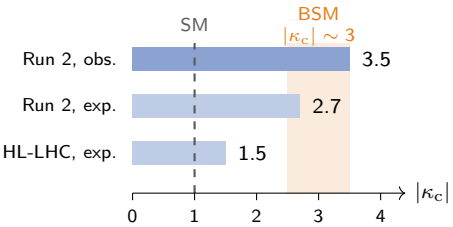
Run 3 starts with higher trigger efficiency and a $t\bar{t}$+$c\bar{c}$ background measured in Run 2

Prospects at the HL-LHC

- ATLAS + CMS input to the European Strategy update
  (arXiv:2504.00672, CMS-HIG-25-002), $3\,\text{ab}^{-1}$ per experiment:
  - $\text{VH}(\text{H} \to \text{c}\bar{\text{c}})$: $\approx 1.6\sigma$ expected
  - **expected** $|\kappa_\text{c}| < 1.5$ at 95% CL: 14 TeV, $\text{VH}(\text{H} \to \text{c}\bar{\text{c}})$ extrapolated with improved c tagging (ATLAS study, assumed for both experiments)
  - 30% better than the 2019 Yellow Report projection
- Next gains: Run 3 data, ML tagging, detector upgrades; theory modelling and tagger calibration become limiting

κ_c at $\mathcal{O}(1)$: first sensitivity to whether the Higgs gives charm its mass

Direct searches: allowed $|\kappa_\text{c}|$ at 95% CL



Run 2: $\text{t}\bar{\text{t}}\text{H}$+VH (PRL 136 (2026) 011801); HL-LHC: ATLAS+CMS, $3\,\text{ab}^{-1}$ each. Every bar still includes $\kappa_\text{c} = 0$.

Summary

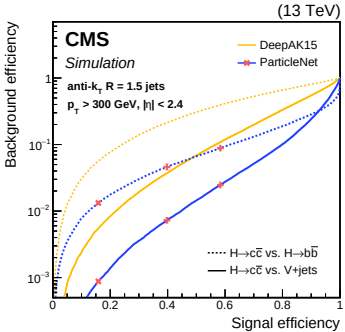
- The charm Yukawa is the most accessible 2nd-generation quark coupling; CMS probes it in decay, in production and indirectly
- $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ + $\mathrm{VH}(\mathrm{H} \to \mathrm{c\bar{c}})$: $|\kappa_\mathrm{c}| < 3.5\,(2.7)$, the most stringent direct constraint and the best expected sensitivity
- H+c production: $|\kappa_\mathrm{c}| < 47\,(51)$, independent of the $\mathrm{H} \to \mathrm{c\bar{c}}$ decay
- Off-shell + on-shell $\mathrm{H} \to 4\ell$: $\kappa_\mathrm{c} \in [-3.3, 2.8]$ with $|\kappa_V| \leq 1$ (about ± 8 without)
- All κ_c results are Run 2; Run 3 searches start with higher trigger efficiency and a $\mathrm{t\bar{t}}$+$\mathrm{c\bar{c}}$ background already measured in Run 2
- **Does the Higgs give charm its mass?** Today $|\kappa_\mathrm{c}| \lesssim 3$–$3.5$, entering BSM-motivated territory; HL-LHC expected $|\kappa_\mathrm{c}| < 1.5$: approaching, not yet reaching, sensitivity to $\kappa_\mathrm{c} = 0$

Thank you!

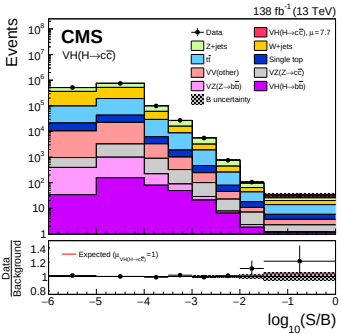
Backup

VH($\mathrm{H} \to \mathrm{c\bar{c}}$): event categories and sensitivity

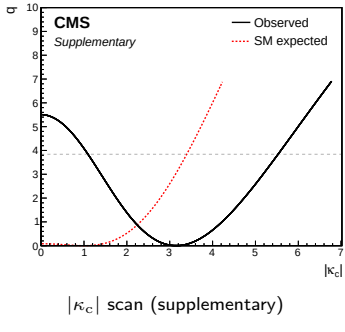
PRL 131 (2023) 061801



ParticleNet merged-jet $\mathrm{H} \to \mathrm{c\bar{c}}$ tagging



Events vs. $\log_{10}(S/B)$



$|\kappa_c|$ scan (supplementary)

Best fit $\mu = 7.7^{+3.8}_{-3.5}$; merged $\mu = 8.7^{+4.6}_{-4.0}$, resolved $\mu = -9.5 \pm 9.6$. $\sigma\mathcal{B} < 0.94$ (0.50) pb.
μ limits with CLs; κ_c intervals from profile-likelihood scans.

The upward fluctuation comes from the merged-jet category

VH($\mathrm{H} \to \mathrm{c\bar{c}}$): analysis details

PRL 131 (2023) 061801; strategy as in JHEP 03 (2020) 131

- **Two topologies, orthogonal**: H candidate from a large-R ($R = 1.5$) jet with p_{T} above (merged) or below (resolved) 300 GeV; same lepton channels
- **Resolved**: DeepJet CvsL/CvsB; the two highest-CvsL jets form the H candidate; leading-jet cuts at ≈40% c efficiency (≈4% light, ≈16% b mistag). c-jet energy regression, FSR-jet recovery and a 2L kinematic fit improve the $m(\mathrm{H_{cand}})$ resolution by up to 20%. BDT per channel, fit to the BDT score
- **Merged**: ParticleNet $\mathrm{c\bar{c}}$ tagger, 4–7× better rejection than DeepAK15; 3 working points (≈58, 40, 16% $\mathrm{c\bar{c}}$ efficiency) define 3 purity categories. ParticleNet mass regression: ≈50% better mass resolution. BDTs decorrelated from $m(\mathrm{H_{cand}})$; fit to $m(\mathrm{H_{cand}})$
- **Backgrounds**: V+jets and $\mathrm{t\bar{t}}$ normalised in dedicated control regions, following the 2016 analysis (V+jets split by jet flavour, free rate parameters)
- **VZ(Z→ $\mathrm{c\bar{c}}$) validation**: same analysis, resolved BDTs retrained with VZ as signal, merged unchanged. $\mu_{\mathrm{VZ}} = 1.01^{+0.23}_{-0.21}$; 4.4$\sigma$ merged, 3.1σ resolved, 5.7σ combined (exp. 4.7, 3.3, 5.9)

Uncertainty breakdown

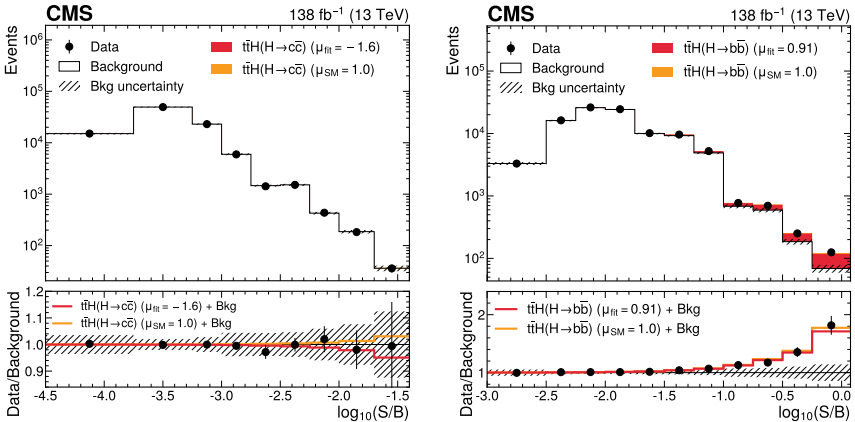
Source	$\Delta\mu/\Delta\mu_{\mathrm{tot}}$
Statistical	**85%**
Bkg. normalisations	37%
Experimental	48%
Simulated sample sizes	37%
c-jet identification	23%
Jet energy scale, resolution	15%
Simulation modelling	11%
Luminosity	6%
Lepton identification	4%
Theory	22%
Backgrounds	17%
Signal	15%

For $\hat{\mu} = 7.7^{+3.8}_{-3.5}$; simulation size driven by negative weights in NLO V+jets

Statistics-limited; the c-tagging and simulation-size systematics come next

$t\bar{t}H(H \to c\bar{c})$: event yields

PRL 136 (2026) 011801



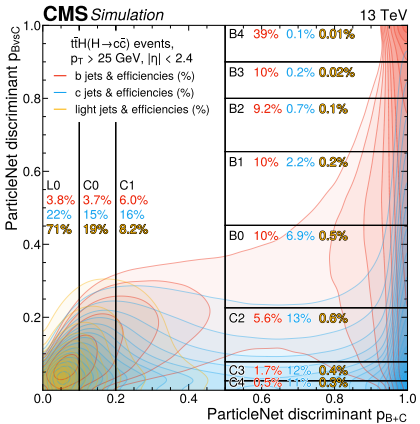
Event yields vs. $\log_{10}(S/B)$; the $t\bar{t}H(c\bar{c})$ panel shows the published best fit $\hat{\mu} = -1.6 \pm 4.5$

$t\bar{t}H(c\bar{c})$ best fit $\hat{\mu} = -1.6 \pm 4.5$: compatible with zero and with the SM

$t\bar{t}H(H \to c\bar{c})$: flavour tagging and by-products

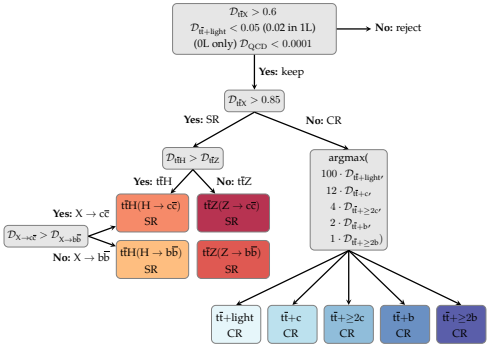
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- Each jet gets ParticleNet scores; the (b+c vs. light, b vs. c) plane is split into 2D categories
- ParT event classifier separates $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z$, $t\bar{t}$+HF
- $t\bar{t}Z(Z \to c\bar{c})$ as a standard candle: $\mu = 1.0 \pm 0.8$
- By-product: $t\bar{t}H(H \to b\bar{b})$ $\mu = 0.91^{+0.26}_{-0.22}$, 4.4σ (4.5 exp.)

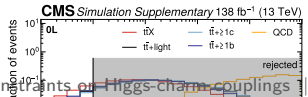


b, c, light jets in the ParticleNet (b+c vs. light, b vs. c) plane and the 2D tagging categories

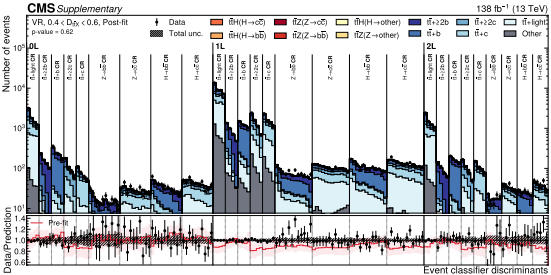
$t\bar{t}H(H \to c\bar{c})$: event categorisation and validation



Categorisation: $D_{t\bar{t}X} > 0.85$ goes to an SR by $D_{t\bar{t}H}$ vs. $D_{t\bar{t}Z}$ and $D_{X\to c\bar{c}}$ vs. $D_{X\to b\bar{b}}$; 0.6–0.85 to the CR of the largest weighted $t\bar{t}$+jets score (weights 100, 12, 4, 2, 1)



PRL 136 (2026) 011801 (supplementary)

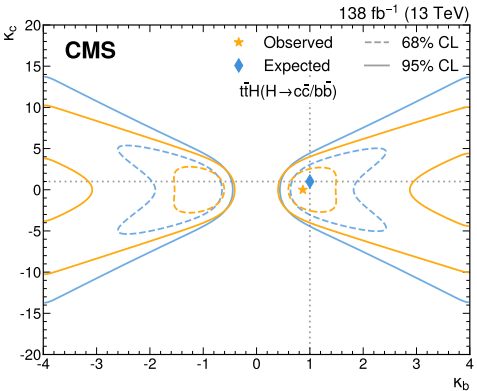


Validation region $0.4 < D_{t\bar{t}X} < 0.6$, split into CR-like (0.40–0.58) and SR-like (0.58–0.60) parts with background purities matched to the analysis region

- c-enriched regions ($t\bar{t}H(c\bar{c})$, $t\bar{t}Z(c\bar{c})$, $t\bar{t}+\geq$2c) need $\geq$2 c jets, $t\bar{t}+$c needs $\geq$1
- Fit variable: the ParT discriminant of each category; $t\bar{t}+$b, $+\geq$2b from a dedicated 4FS $t\bar{t}b\bar{b}$ sample

$t\bar{t}H(H \to c\bar{c}/H \to b\bar{b})$: (κ_b, κ_c) plane

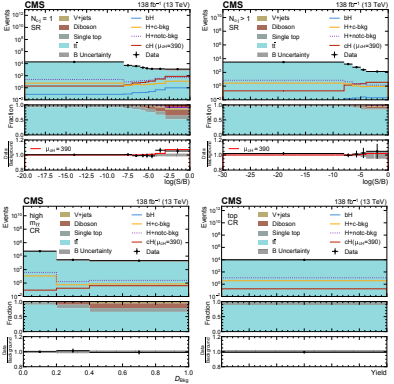
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orange: observed, blue: expected; dashed 68%, solid 95% CL

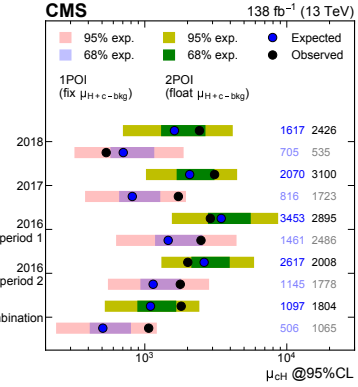
- Simultaneous $t\bar{t}H(H \to b\bar{b})$ + $t\bar{t}H(H \to c\bar{c})$ fit constrains both Yukawas
- Signal strengths: $t\bar{t}H(b\bar{b})$ $0.91^{+0.26}_{-0.22}$; $t\bar{t}H(c\bar{c})$ -1.6 ± 4.5; $t\bar{t}Z(b\bar{b})$ $1.47^{+0.45}_{-0.41}$; $t\bar{t}Z(c\bar{c})$ $1.02^{+0.79}_{-0.84}$

cH(H→WW): post-fit distributions and per-period limits



1-POI post-fit distributions of the BDT discriminants

JHEP 08 (2026) 051



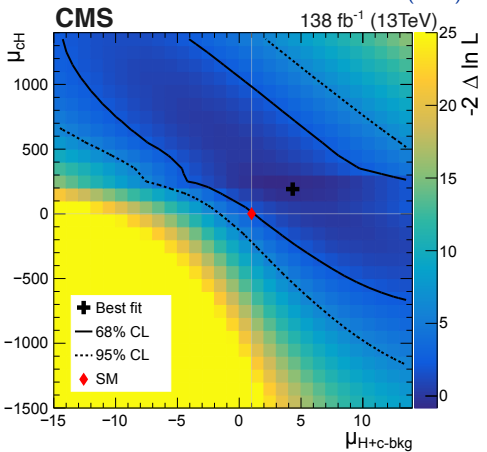
μ_{cH} limits per data-taking period, 1 and 2 POI

Best fit (1 POI): $\mu_{cH} = 390^{+380}_{-270}$, within 1.5σ of the SM; statistics-limited, then flavour-scheme and ggH HF modelling.

cH(H→WW): floating the H+c background

- 1 POI: only μ_{cH} floats; the κ_{c}-independent H+c background (from ggH etc.) is fixed to its SM prediction
- 2 POI: H+c background also floats; correlation with cH: -87%
- 2-POI limit: $\mu_{\mathrm{cH}} < 1804$ (1097)
- The combination with $\gamma\gamma$ uses the 1-POI fit

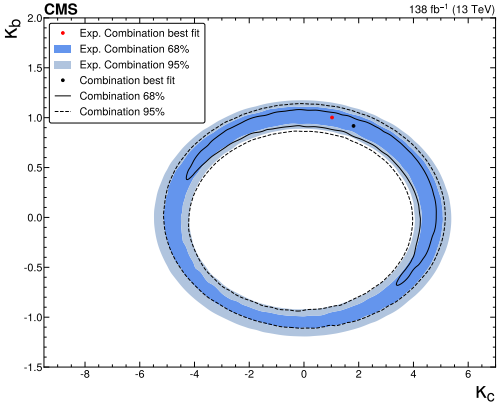
JHEP 08 (2026) 051



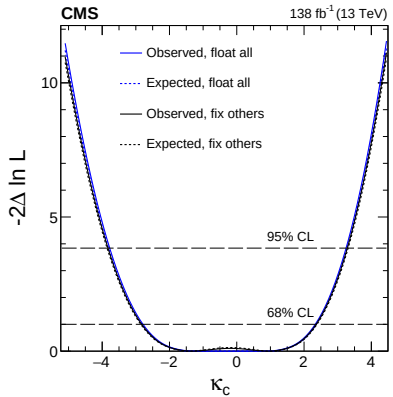
WW: μ_{cH} vs. $\mu_{\mathrm{H+c\ bkg}}$

Floating the H+c background weakens the WW limit from 1065 to 1804

Indirect: Higgs p_{T} spectrum and on-shell H$\rightarrow 4\ell$



JHEP 04 (2026) 093: (κ_{b}, κ_{c}) from the Run 2 $p_{\mathrm{T}}^{\mathrm{H}}$ combination ($\gamma\gamma$, ZZ, resolved $\tau\tau$), $\mathcal{B}$ coupling-dependent; 2D contours only

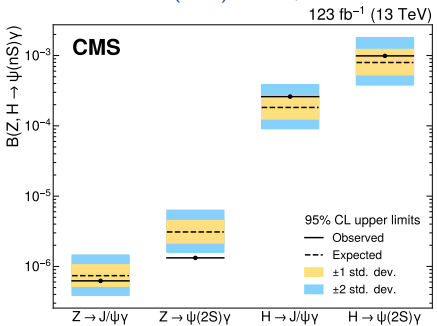


PRD 112 (2025) 112001 (γH + on-shell H$\rightarrow 4\ell$): κ_{c} scan; solid obs., dashed exp.

Rare exclusive decays: H→J/$\psi\gamma$, ψ(2S)γ

- Direct ($\propto \kappa_c$) and indirect (H→ $\gamma\gamma^* \to \gamma$ J/ψ) amplitudes interfere ⇒ sign sensitivity
- 123 fb^{-1}: $\mathcal{B}(\mathrm{H} \to \mathrm{J}/\psi\gamma) < 2.6 \times 10^{-4}$, $\mathcal{B}(\mathrm{H} \to \psi(2\mathrm{S})\gamma) < 9.9 \times 10^{-4}$
- 95% CL: $\kappa_c/\kappa_\gamma \in (-157, 199)$; with $\kappa_\gamma = 1$: $\kappa_c \in (-166, 208)$, exp. $(-128, 169)$

PLB 865 (2025) 139462, CMS-SMP-22-012

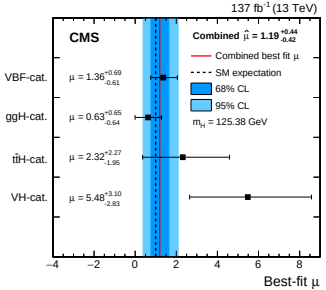


95% CL upper limits on $\mathcal{B}$(Z, H→ $\psi(n\mathrm{S})\gamma$), obs. and exp.

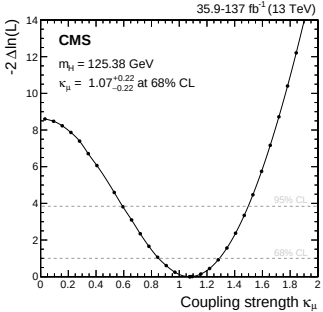
H$\to \mu\mu$: evidence for a 2nd-generation Yukawa

JHEP 01 (2021) 148, CMS-HIG-19-006

- 137 fb^{-1}, four categories: ggH, VBF, VH, $\mathrm{t\bar{t}H}$
- Excess of **3.0**σ (2.5σ exp.): first evidence for H decays to 2nd-generation fermions
- $\mu = 1.19^{+0.40}_{-0.39}\,(\mathrm{stat})^{+0.15}_{-0.14}\,(\mathrm{syst})$
- $\kappa_\mu = 1.07$; 95% CL: $0.59 < \kappa_\mu < 1.50$ (6 free κ, SM total width)
- Rare decays incl. $\mu\mu$: L. D'Eramo's plenary (Wed)



$\hat{\mu}$ per production category and combined

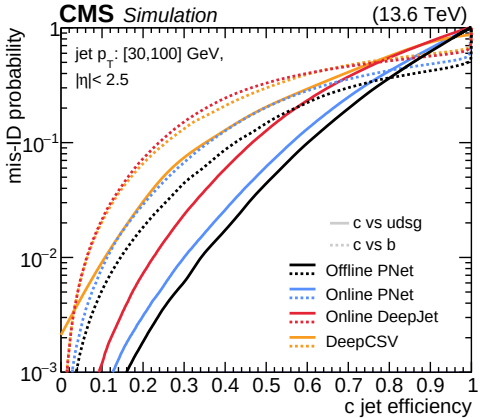


$-2\Delta \ln L$ vs. κ_μ

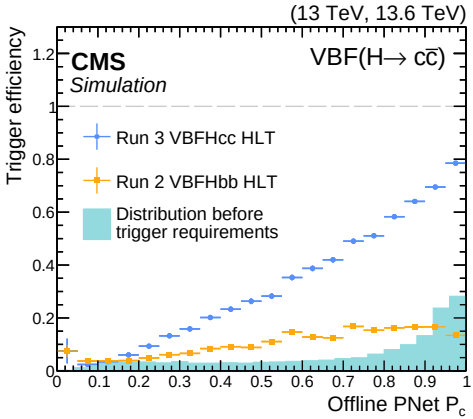
The muon Yukawa is SM-like within about 20% (68% CL); charm is the next 2nd-generation test

Run 3 HLT flavour tagging

arXiv:2608.18331, CMS-BTV-25-002



Simulation, jet p_T 30–100 GeV: light (solid) and b (dashed) misidentification vs. c-jet efficiency, online vs. offline taggers



Simulation: VBF H→ $c\bar{c}$ trigger efficiency vs. offline ParticleNet c score, Run 3 VBF H→ $c\bar{c}$ vs. Run 2 VBF H→ $b\bar{b}$ trigger

Likely questions

- **How large is the VH excess?** Best fit $\mu_{\mathrm{VH}} = 7.7^{+3.8}_{-3.5}$, about 1.9σ above $\mu = 1$ (estimated from the quoted uncertainty; the paper calls it a small upward fluctuation). This is why VH alone excludes $|\kappa_{\mathrm{c}}| < 1.1$ and why $\mathrm{t\bar{t}H}$ alone (3.0) beats the combination (3.5) on observed
- **What limits the direct searches?** Statistics first. $\mathrm{t\bar{t}H(c\bar{c})}$: statistical 74% of the total uncertainty, then $\mathrm{t\bar{t}}$+c and $\mathrm{t\bar{t}}$+light modelling (≈32%, ≈29%). VH($\mathrm{c\bar{c}}$): simulation sample size (≈37%), c-tagging efficiency (≈23%), theory (≈22%)
- **And ATLAS?** VBF + VH($\mathrm{H \to c\bar{c}}$) (arXiv:2511.21911): $|\kappa_{\mathrm{c}}| < 4.7$ (3.9 exp.), same assumptions as CMS (other couplings SM, no BSM decays); its VBF part includes 51.5 fb^{-1} of Run 3 data
- **Why is cH an order of magnitude weaker than $\mathrm{H \to c\bar{c}}$?** cH $\sigma\mathcal{B}$ is tiny (≈0.2 fb in $\gamma\gamma$) and ggH produces the same H+c final state independently of κ_{c}; in $\mathrm{H \to c\bar{c}}$, κ_{c} changes the branching fraction of Higgs bosons from much larger production modes (VH, $\mathrm{t\bar{t}H}$)
- **Why is cH($\gamma\gamma$) obs. (38) much better than exp. (73)?** The best fit is negative, $\hat{\mu}_{\mathrm{cH}} = -230^{+170}_{-251}$
- **Why no κ_{c} from boosted $\mathrm{H \to c\bar{c}}$?** The paper quotes only μ; the result is ggH-dominated
- **How much of the Run 3 trigger gain is the tagger?** Part of it is rate budget: 10 Hz in 2018 vs. 60, 115, 170 Hz in 2022, 2023, 2024
- **Are the H→ 4ℓ intervals sign-sensitive?** Not really: the small asymmetry comes from charm-loop interference in ggH and H→gg; the constraint is essentially on κ_{c}^2

References

- VH($\mathrm{H} \to \mathrm{c\bar{c}}$): CMS, PRL 131 (2023) 061801, arXiv:2205.05550 (HIG-21-008)
- $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ + VH combination: CMS, PRL 136 (2026) 011801, arXiv:2509.22535 (HIG-24-018)
- VH($\mathrm{H} \to \mathrm{c\bar{c}}$) with 2016 data (analysis strategy): CMS, JHEP 03 (2020) 131, arXiv:1912.01662 (HIG-18-031)
- Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$: CMS, PRL 131 (2023) 041801, arXiv:2211.14181 (HIG-21-012)
- cH(H$\to \gamma\gamma$): CMS, JHEP 11 (2025) 060, arXiv:2503.08797 (HIG-23-010)
- cH(H$\to$WW) and combination: CMS, JHEP 08 (2026) 051, arXiv:2508.14988 (HIG-24-009)
- Differential combination Run 2: CMS, JHEP 04 (2026) 093, arXiv:2504.13081 (HIG-23-013)
- γH + H$\to 4\ell$ light Yukawas: CMS, PRD 112 (2025) 112001, arXiv:2502.05665 (HIG-23-011)
- Off-shell + on-shell H$\to 4\ell$: CMS, arXiv:2607.23352, submitted to PRD (HIG-25-003)
- H$\to$J/$\psi\gamma$: CMS, PLB 865 (2025) 139462, arXiv:2411.15000 (SMP-22-012)
- H$\to \mu\mu$: CMS, JHEP 01 (2021) 148, arXiv:2009.04363 (HIG-19-006)
- HLT flavour tagging: CMS, arXiv:2608.18331 (BTV-25-002)
- $\mathrm{t\bar{t}}$+b/c cross sections: CMS-PAS-TOP-26-003
- ATLAS VBF + VH($\mathrm{H} \to \mathrm{c\bar{c}}$): ATLAS, arXiv:2511.21911
- HL-LHC: ATLAS and CMS, arXiv:2504.00672 (HIG-25-002)
- Higgs combination: CMS, Rep. Prog. Phys. 89 (2026) 087801, arXiv:2602.18611 (HIG-21-018)
- Spontaneous flavour violation: arXiv:2410.05236
- HL-LHC Yellow Report: M. Cepeda et al., CERN Yellow Rep. Monogr. 7 (2019) 221, arXiv:1902.00134